Different Fields Have the Same Scaled Ordered K_0: A Disproof of Conjecture 00000006291

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Abstract

The assertion that the zeroth K-group completely classifies regular rings fails already for the fields $\mathbb{F}_2$ and $\mathbb{F}_3$. Both are von Neumann regular, and their scaled ordered K_0 groups are isomorphic to $(\mathbb{Z}, \mathbb{Z}_{\geq 0}, 1)$, although the rings are not isomorphic. The Lean proof constructs the Grothendieck group from actual finite-dimensional vector spaces and linear equivalences, verifies its universal property, and proves the counterexample with the positive cone and the unit class retained.

Reading of the source. Both language versions assert that the zeroth group is a complete invariant of regular rings and describe a classification by ordered groups. We use the standard unital von Neumann meaning of regular:

$$\forall a \in R\ \exists b \in R, \qquad aba = a.$$

The zeroth group is the algebraic Grothendieck group $K_0(R)$ of finitely generated projective right R-modules under direct sum. Its positive cone consists of the classes of actual modules, and its distinguished order unit is $[R]$. The rings here are commutative, so right versus left is immaterial. Completeness for ring isomorphism would imply, in particular,

$$(K_0(F), K_0(F)^+, [F]) \cong (K_0(E), K_0(E)^+, [E]) \quad \Longrightarrow \quad F \cong E \tag{1}$$

for every pair of fields F, E, since fields are regular. We disprove this necessary restriction. Retaining the positive cone and unit makes the counterexample stronger than one involving only abstract abelian groups. No common ground field or narrower class of regular rings is stipulated in the source; a classification theorem with such extra restrictions is outside the claim refuted here.

Theorem 1. *The rings $\mathbb{F}_2$ and $\mathbb{F}_3$ are von Neumann regular and nonisomorphic, but*

$$(K_0(\mathbb{F}_2), K_0(\mathbb{F}_2)^+, [\mathbb{F}_2]) \cong (\mathbb{Z}, \mathbb{Z}_{\geq 0}, 1) \cong (K_0(\mathbb{F}_3), K_0(\mathbb{F}_3)^+, [\mathbb{F}_3]).$$

Consequently (1)*, and hence the stated complete-invariant assertion for regular rings, is false.*

Lemma 1. *For every field F, dimension induces a scaled ordered group isomorphism*

$$(K_0(F), K_0(F)^+, [F]) \cong (\mathbb{Z}, \mathbb{Z}_{\geq 0}, 1).$$

Proof. Every finitely generated F-module is a finite-dimensional vector space and has a finite basis. A free module is projective: to lift a linear map through a surjection, choose preimages of the images of basis vectors and extend linearly. Thus the finitely generated projective modules over F are precisely its finite-dimensional vector spaces. Their isomorphism classes form a commutative monoid under finite direct sum, which can be realized as the Cartesian product with componentwise vector operations. Dimension identifies this monoid with $(\mathbb{N}, +)$, because

$$P \cong F^{\dim_F P}, \qquad P \cong Q \iff \dim_F P = \dim_F Q, \qquad \dim_F(P \oplus Q) = \dim_F P + \dim_F Q.$$

The following explicit group completion also specifies the construction used in Lean.

Take pairs (P, Q) of actual finite-dimensional vector spaces. Declare

$$(P, Q) \sim (P', Q') \quad \Longleftrightarrow \quad P \oplus Q' \cong P' \oplus Q. \tag{2}$$

This is equivalent to equality of the integer differences $\dim P - \dim Q$ and $\dim P' - \dim Q'$ and is therefore an equivalence relation. It also agrees with the customary stabilized relation allowing an auxiliary summand T on both sides of the isomorphism in (2): equality of the resulting dimensions cancels $\dim T$, and the converse follows by taking $T = 0$.

On equivalence classes put

$$[(P, Q)] + [(P', Q')] = [(P \oplus P', Q \oplus Q')], \qquad -[(P, Q)] = [(Q, P)].$$

These operations are well-defined by the dimension criterion. The map

$$d([(P, Q)]) = \dim_F P - \dim_F Q$$

is well-defined and injective by (2), and is surjective because an integer z is represented by $(F^{\max(z,0)}, F^{\max(-z,0)})$. It is an additive isomorphism to $\mathbb{Z}$.

For completeness, this quotient has the defining universal property of the Grothendieck group. Let A be any abelian group and let f assign an element of A to each finite-dimensional vector space, invariant under linear isomorphism, additive under direct sum, and zero on the zero space. Then

$$[(P, Q)] \longmapsto f(P) - f(Q)$$

is well-defined: an isomorphism in (2) gives $f(P) + f(Q') = f(P') + f(Q)$. This map is additive and sends $[(P, 0)]$ to $f(P)$. It is unique, because every class equals $[(P, 0)] - [(Q, 0)]$. Hence the quotient is the Grothendieck group of the specified monoid, namely $K_0(F)$. Equivalently, the exact-sequence presentation gives the same group: short exact sequences of projective modules split, and their relations are exactly direct-sum relations.

The positive cone is the set of classes $[(P, 0)]$. Their images under d are precisely the non-negative integers, since $\dim_F P \geq 0$ and every $n \geq 0$ occurs for $P = F^n$. Thus the order is exactly

$$x \leq y \quad \Longleftrightarrow \quad y - x \in K_0(F)^+ \quad \Longleftrightarrow \quad d(x) \leq d(y).$$

Finally $[F]$ has dimension one. This proves the scaled ordered assertion. □

Proof of Theorem 1. In any field, take $b = a^{-1}$ when $a \neq 0$ and $b = 0$ when $a = 0$; then $aba = a$. Thus both fields are von Neumann regular. Apply Lemma 1 to the two fields and compose one dimension isomorphism with the inverse of the other. The composition preserves addition, order, and the distinguished unit. A ring isomorphism between $\mathbb{F}_2$ and $\mathbb{F}_3$ would be a bijection between their underlying sets, which have different cardinalities 2 and 3. No such isomorphism exists. □

Formal verification and scope

The self-contained file `lean/Main.lean` uses Lean 4.19.0 and Mathlib at commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

Its bundled objects carry actual types, additive groups, scalar actions, and Mathlib's finite-generation instances. The following bridges are proved in Lean:

- Every bundled object is finitely generated and projective, every finitely generated module can be bundled, and every object is linearly isomorphic to the standard finite-dimensional space of its dimension.

- The defining relation is a genuine linear equivalence between cross direct sums; allowing an auxiliary stabilizing module gives the same relation.
- The rank map on the quotient is bijective. The abelian-group structure is transported along it, then proved to be direct sum on pair representatives, with coordinate exchange as additive inverse.
- The group completion has its universal property for every abelian target group. Its partial order is compatible with addition and is proved to be exactly the order coming from actual module classes.
- The unit class equals the class of the field as a module over itself. For every pair of fields an ordered additive group isomorphism preserving this unit is constructed.
- The two `ZMod` fields are regular, no ring isomorphism exists between them, and the universally quantified field-completeness assertion is negated.

The final theorem is

`Conjecture6291.conjecture_6291_false`.

Its type is the negation of `FieldCompleteness`, which quantifies over every pair of fields and assumes regularity and an isomorphism of their scaled ordered groups before demanding a ring isomorphism. This is an explicitly stated necessary restriction of the source's complete-invariant assertion, not a formal construction of K_0 for arbitrary rings. The finite-projective/direct-sum definition is formalized with its universal property; the equivalence with the short-exact-sequence presentation is explained on paper. The remaining unspecified claims concerning continuous geometry are unnecessary once completeness has failed and are not asserted here. The axiom audits use only `propext`, `Classical.choice`, and `Quot.sound`.

Source and reproduction

The exact bilingual statement is included as `SOURCE.md`, from upstream commit

`b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`.

Its source hash and retrieval record are in

`verification/SOURCE_PROVENANCE.md`.

The computation $K_0(F) \cong \mathbb{Z}$ by dimension is also recorded in the Stacks Project, Section 10.55, Example 10.55.2: `stacks.math.columbia.edu/tag/00JC`. The proof above supplies the group, cone, and unit calculation directly.

With the pinned dependencies available, run in the `lean` directory:

```
lake env lean -DwarningAsError=true Main.lean
```

and `lake build` for a fresh project build. Compile `main.tex` with Tectonic or a compatible LaTeX installation. This submission was developed by a delegated AI agent with a self-review; the coordinating agent reviews it separately. No external independent review is claimed.